



Real-World Asset Identity Authentication in Blockchain Enabled Inter-Organizational Process-Aware Systems Involving Adjustable Challenge-Response Evaluation Sets

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Abstract

Blockchain technologies enable transparent and trusting inter-organizational business collaborations in many domains, including also real-world assets (RWA). Consequently, decentralized application (DApp) platforms emerge that record collaboration-relevant data in an immutably traceable way. Such DApps comprise smart contracts that automate the enforcement of defined business protocols and conditions in an environment without reliance on trusted centralized third parties. Accurate RWA identification must include the correlated parties together with their respective roles in public network inter-organizational collaborations (IoC) involving blockchain technologies. Still, enterprises tend to prefer permissioned networks for DApps due to privacy concerns and challenges related to user management. RWA related organizations and involved roles are exposed to contextual changes that demand secure, adaptable, and versatile authentication means before involving them in blockchain-enabled IoC processes. Given these factors, the paper explores the combination of the Logistics BDT DApp for RWA tracking together with the multifactor challenge set self-sovereign identity authentication (MFSSIA) for effective counterfeit circulation prevention. This combination comprises non-fungible token (NFT)-based RWA identity authentication as a running case. This running case involves a set of configurable challenges and integrates them into the existing DApp combination to manage trustable IoC processes.

Keywords Blockchain · RWA · Multi-factor · Challenge set · Identity · Authentication · Non-fungible token · Decentralized application

Introduction

The integration of blockchain technology in the implementation of interorganizational collaboration (IoC) has recently increased. IoC systems benefit from blockchain technologies in that the latter are decentralized, immutable, and transparent [2, 15]. Briefly, a blockchain is a decentralized ledger of blocks in which stakeholders contribute data through a

formalized and cryptographically secure consensus mechanism. The latter ensures that the data recorded on the chain remain accurate and consistent [45].

Several research studies have also presented numerous applications of IoC on blockchains, such as supply chains for various products and services in sectors such as agriculture, healthcare, construction, and more. Consequently, numerous applications of blockchain technologies in IoC are presented in research publications, such as the following various research works [2], agriculture [33], healthcare [37], construction [35], luxury [20] etc. Note that each example requires some form of RWA-identity authentication.

There is a well-known need to incorporate flexible and personalized identity mechanisms to authenticate the legitimacy of organizations and parties within an organization while implementing IoC on public blockchains. Briefly, public networks allow unrestricted access to the network, while permissioned variants are private networks that restrict access to specific known parties [45]. This is

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important because organizations are often dynamic, with people constantly changing and their roles evolving. In some instances [30], when configuring blockchain applications, certain functions and roles are associated or hardcoded with specific keys during the deployment and instantiation of the contract, especially on public blockchains. The keys serve as identification for users who are associated with particular organizations and are assigned specific duties during inter-organizational collaborations on the blockchain. As stated by Sheth (2019) [45], smart contracts, which are software programs running on a blockchain, have immutability. Note that it is impractical to modify the code on-chain to adjust to changes within an organization.

The research results suggest diverse approaches to authenticate RWA and regulate user access to specific related functions of public blockchain organizations. For example, role-based access control (RBAC) systems are considered to manage and restrict user access to specific functions in blockchain DApps [18, 40]. The evolving understanding of self-sovereign identity (SSI) [7] includes decentralized identifiers (DID) that comprise cryptographically signed verifiable credentials (VC) [36, 43]. A recent systematic review of the literature [22] identifies key challenges in ensuring that smart contract development is user-centered, on the one hand, and also legally compliant, on the other. The systematic literature review proposes a framework for integrating user requirements and the legal side into one development process. The characteristic of SSI is the available digital identity management model with which individuals and parties of an organization have control over their own personal identity data, including how their respective data are shared and utilized [7]. For blockchain DApps, the verification of users and their associated attributes can be achieved with a token-based system [34, 50]. In [39], the authors present an SSI system of self-sovereign identity authentication multi-factor challenge set (MFSSIA) that comprises contextually configurable challenge sets for the identity authentication of devices, systems, organizations, and individual persons. Note that MFSSIA has been extensively studied in previous research papers [38, 39] that focus on a formal investigation of challenge-response lifecycle management. As a DApp, MFSSIA is implemented as a challenge-response system, in which oracles retrieve off-chain facts to validate responses and authenticate users. Note that the literature review in the sequel of this paper provides a further detailed discussion of these research works, discussing their respective contributions to identity authentications on public blockchains.

We discover that current research is lacking on how to configure and automate challenge-response lifecycle management systems to identity authenticate devices, systems, organizations, and the individual involved in IoC processes on public blockchains. An application of MFSSIA in IoC improves the security of and trust in the exchange

of sensitive data related to RWA. Consequently, this paper poses the main research question: How to design a configurable challenge-response lifecycle management mechanism for the identity authentication of organizations and individual persons involved in executing IoC on blockchains? To establish a separation of concerns, the main research question is used to derive the following research questions. What is the current process depicting a decentralized application for executing IoC on blockchains? What are the useful sets of challenges for verifying parties within organizations? What are the necessary integration steps of challenges sets for verifying parties in organizations when executing IoC on blockchains?

The remainder of this paper is organized as follows. First, Sect. [Research Method and Related Work](#) introduces the research method we instantiate for this paper together with related work. Next, Sect. [Preliminaries](#) presents a running case and background information related to this study. Section [IoC Context Luxury RWA Identity Authentication](#) addresses the first research question by providing a description of a sample business process of an existing DApp that facilitates organizational collaboration on a blockchain. Section [Challenge-sets for securing parties within organizations](#) answers the second research question by proposing a set of challenges that can improve the authentication of parties involved in the described business collaboration. Section [NFT-Based Luxury RWA Authentication System Challenge-Set Integration](#) addresses the third research question by modifying the collaborative process through the integration of configurable challenge sets and blockchain oracles, in order to improve user authentication in blockchain-based collaborative processes. Section [Evaluation and Discussion](#) presents the evaluation of this study, including an assessment of existing blockchain technology stacks to implement the proposed system and a comparative analysis of the authentication system described in this study and other relevant works.

Research Method and Related Work

In addition to presenting the adopted research method in Sect. [Research Method](#), we also discuss related work in Sect. [Related Literature Review](#).

Research Method

The study presented in this article adopts the design science research methodology (DSR), which offers a systematic approach to the development and evaluation of information system artifacts (IS) [48]. This framework includes techniques to generate new theories in the form of constructs, models, and methods. The main objective of this research

is to propose a method for identifying the requirements of collaborative systems on the blockchain.

We show the application and instantiation of the DSR method in Fig. 1 according to a study adapted from [48], comprising the following three aspects: environment, knowledge base, and IS research. Although the environment and knowledge base are essential to effectively execute DSR, the environment column specifically shows the importance of this research in addressing people, organizational issues, and technological aspects. In the case of this paper, the focus is on the use of blockchains for the design and development of DApps. For the organizational approach, this research employs IoC processes, and the problem of modeling the integration of several DApps is addressed. Consequently, in Fig. 1, the RWA buyers and their sellers are the people roles in the running case.

The organizations involved are the RWA management organization and the one acting as an RWA buyer that follows cross-organizational trade processes. The organizations shown in Fig. 1 are the RWA trader selling the RWA, the owner organization of the trade process, and the RWA buying organization in the case of a business-to-business (B2B) situation. The adopted running case must be relevant because there exists a connection with observable events in reality to meet the business needs collected as input for the IS research column in Fig. 1. Finally, the knowledge base on the right of Fig. 1 represents the scientific models, methods, frameworks, and validation criteria that are applied to the IS research column in the center to achieve a high degree of rigor. We discuss and explain the foundations and methodologies in the sequel of this paper. Briefly, the foundations listed in Fig. 1 are MFSSIA, non-fungible tokens (NFT) [42], decentralized knowledge graphs (DKG) [49], blockchain oracles, the business process model and notation (BPMN) [16], and decentralized identifiers (DID) [28]. However, the methodologies listed are used to develop a secure approach that authenticates the identity of NFTs. With this corpus of knowledge base, the assessments are performed in the evaluation section of this paper for related

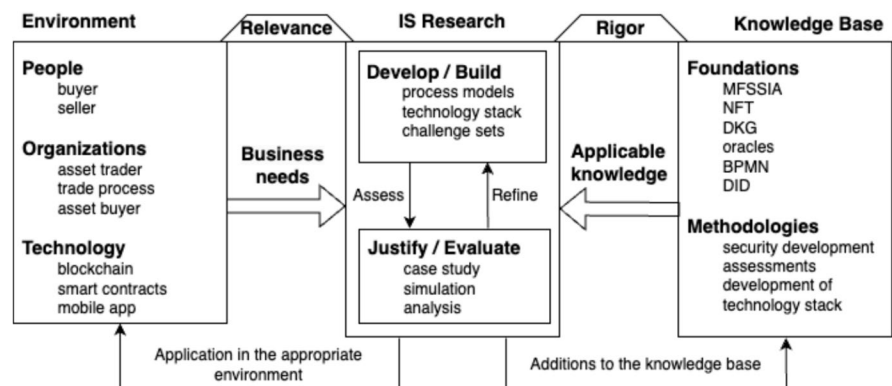
other authentication approaches. For IS research in the center of Fig. 1, the center column shows the artifacts developed being the process models, a technology stack and a set of challenges. We consider the adopted running case for artifact evaluations, the simulation of generated IoC processes, and an analysis of the suitable set of technologies, challenges, and security needs for ensuring NFT identity authentication. The artifacts are refined on the basis of the results of the evaluation to achieve a high degree of rigor and relevance of the IS research. Finally, Fig. 1 shows that the generated IS research results are applicable to the running case and simultaneously also generate additions to the knowledge base of the scientific state of the art.

Related Literature Review

For information systems, the application of access control mechanisms is a means of limiting function and data access. The authors in [34] explore a flexible token-based mechanism that enforces access control on resources in DApps. A smart contract uses a token service to grant or deny access to a client who wishes to access a smart contract function. The tokens are classified into various permission levels, thus determining the type of resources or functions that can be accessed by the client users.

The RBAC model is another approach to managing access to smart contract functions executed on a blockchain. However, these roles are not further configurable and remain immutable once the smart contract application is instantiated. The research work in [1, 18, 40] shows that conventional smart contract applications can also incorporate predefined roles such that access to specific functions is limited to users with the assigned roles. Research [17] extends the RBAC system by addressing the need to apply democratic concepts in the management and modification of user roles in decentralized applications. In the IoC context, the method developed and implemented in [17] ensures that parties who approve certain business process workflows have the required role based on the policy defined in the organization.

Fig. 1 The DSR method application context, adapted from [48]



This is achieved by ensuring that tasks involving administrative decisions have a specified number of approvals by users within the admin role for the task to be executed.

SSI-based authentication schemes provide approaches for managing identities and users when performing functions in blockchain applications. DIDs are frequently used SSI verification mechanisms [12, 43]. DIDs are universal resource identifiers that connect a party or an entity to a document, enabling trustable interaction such that a party can digitally prove ownership of a particular document [43]. VCs provide a standardized data format for cryptographically verifiable digital credentials, such that the credentials are issued by a DID to a party proving ownership of a document or RWA [36]. The DID system defines three main roles: issuer, receiver, and verifier. The issuer creates a cryptographically signed VC for a particular DID that is handed over to a receiver. The verifier checks the digital signature contained in the VC linked to a DID document to verify that the credentials are generated by a particular issuer. The verifier also checks that the signature is still valid and that it has not been revoked. Some privacy issues could arise by applying DID/VC systems for implementing user access control in smart contracts. Yet, this can be addressed using oracles to perform the verifications [51].

Another example of an SSI-based user authentication scheme is the soul-bound token (SBT). Unlike DID-based verifiable credentials, an SBT is a unique non-transferable token used to digitally represent a person or an entity such as a business organization [50]. The SBT features can be used to capture the properties of an entity, such as historical activities or records related to a person or entity. Several SBTs can be linked to a single user so that they provide transparent provenance of the behaviors and reputations associated with the entity. Although SBT can be used to represent organizations such as decentralized autonomous organizations, the current specification does not show how different roles can be assigned to individuals representing the same organization. Theoretically, a multi-party wallet can be used to capture the ownership of SBT. Still, the problem of role configurability remains, as organizations are generally dynamic entities.

The self-configurable challenge-response system has also been used to demonstrate SSI in blockchain applications [32, 38]. The authcoin system described in [32] proposes a challenge-response system that allows two parties to authenticate each other by posing challenges where both parties know the correct responses to the challenges. The result of the response check is stored on a blockchain so that both parties have access to the status of the authentication. The paper [38] shows the MFSSIA implementation that extends the authcoin protocol. The extension is achieved by providing a challenge-response system that allows parties that execute interorganizational processes to check and

verify business conditions before executing a particular business process task. The challenges and correct responses are initially known by both parties, so the correct responses are hashed and stored on a chain. Still, if business conditions change, the parties will have to deploy new challenges and responses. Otherwise, the authentication will not be successful. This ensures that parties executing autonomous inter-organizational processes on a blockchain have a common understanding and overview of the business conditions before they are executed.

However, in permissioned blockchains, such as Hyperledger Fabric (HLF), the user authentication system is based on PKC. Here, we specify the organizational-level certificate authority that issues and revokes user certificates. The certificates can contain information such as the organizational name, units/departments, and user roles. Therefore, at the chain code (smart contract) level, the client identity function can be used to check and limit which users have access to a particular function based on the role and departments listed in the user certificate [5, 29].

Decentralized Identifiers (DIDs) have emerged as a key technology to improve privacy and ensure security in the context of tracking RWAs in logistics and supply chains, verifiable digital identities. DIDs allow entities, whether individuals, organizations, or devices, to possess a unique, self-sovereign identity without relying on a centralized authority. Public-key cryptography enables this autonomy so that entities can prove identity ownership without losing their privacy. In a study about DID [31], the authors explore the Internet of Things (IoT) in logistics and supply chain as application cases. The essence of this research is to evaluate the DID deployment in real-world IoT scenarios to address IoT device constraints, and provide solutions, e.g., proxy approaches for devices that are not able to operate complex cryptographic operations.

To foster a unified cross-platform approach, this study additionally explores how DID improves privacy, highlighting the aspect of interoperable DID technologies aligned with the W3C DID specification.¹ Transcending established identification methods, DID with integrated blockchain technologies introduces a novel degree of decentralized identification in IoT. Thus, by adopting transparency and immutability into DID, novel IoT applications are attainable, e.g., from smart homes to broader industry use cases. The resulting decentralized DIDs mitigate common security threats on the one hand and also provide innovative access control and identity verification solutions on the other hand to enhance the trustworthiness and privacy of digital interactions [27].

¹ <https://www.w3.org/TR/did-core/>.

Preliminaries

The efficient and transparent execution of IoC is pivotal for DApps while posing a considerable challenge at the same time. In this section, we investigate the running-case-relevant IoC processes on blockchain together with current methodologies, their inherent limitations, and the pressing need for robust verification mechanisms within organizations. As a result, this section provides the concepts relevant for this paper to address process modeling, challenge identification, and integration steps required to fill a gap in IoC processes on blockchains.

Thus, Sect. **Running Case** begins by introducing a running case for the paper expressed in the problematic AS-IS state. Next, we provide in Sect. **Blockchain Concepts** the foundational concepts pertaining to blockchain technologies.

Running Case

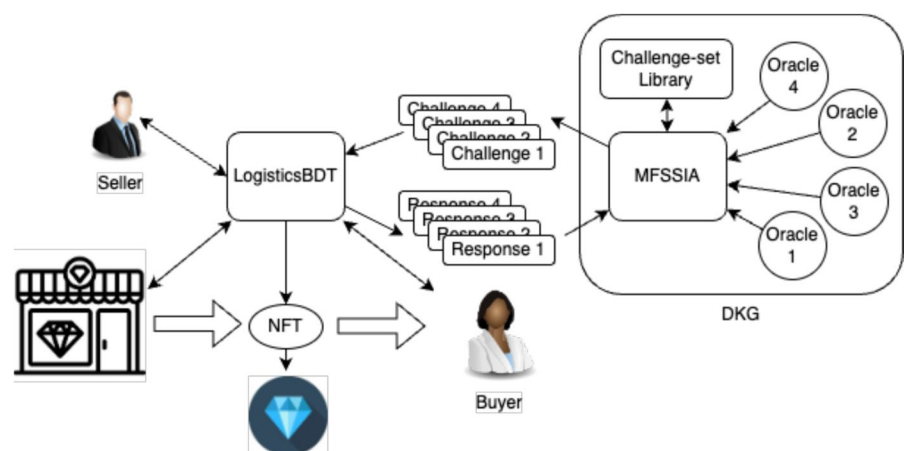
In an earlier research work [47], the authors show the design and implementation of a DApp for the authentication of the luxury category RWA using NFTs. Briefly, an NFT [8] is a blockchain token that is uniquely assigned to an RWA such as media, a piece of art, digital content, etc. Thus, an NFT represents an irrevocable digital ownership certificate of authenticity for a given physical or digital asset. Inspired by our previous work, a running case is examined about NFTs for the identity authentication of luxury RWAs. With digitizing the unique identities of luxury RWAs in NFTs, provenance and authenticity is ensured. The key stakeholders of the running case comprise the authorized retailers of the luxury RWA and the customers, who aim to benefit from blockchain technologies that provide enhanced security and transparency.

As Fig. 2 shows, the process begins by digitally mapping the luxury RWA to an NFT to thus use the different available identifiers, such as the serial number and product ID. In our

running case, Fig. 2 depicts a diamond in a jewelry store that is mapped onto an issued NFT. This mapping creates a digital twin of the physical item on the blockchain, with the NFT's genesis transaction serving as immutable proof of authenticity. Still, to address emerging security challenges and improve trust in this system, the running case proposes to secure the authenticity of an issued NFT taking advantage of the configurable challenge set concept of the MFSSIA [38] framework. This approach aims to fortify the NFT minting process against potential vulnerabilities, ensuring that only authorized entities can create or transfer NFTs corresponding to genuine luxury items. By integrating such security measures, the blockchain application not only protects the interests of all stakeholders, but also sets a new standard for the authentication of high-value RWAs in the digital age. Note that if we assume the running case extends for an RWA along a longer supply chain, MFSSIA is also instrumental for identity authentication of all devices, systems, organizations, and individual persons along each step.

Expanding on the running case of Fig. 2, the depiction shows the LogisticsBDT blockchain DApp [47] for NFT lifecycle management issuing an RWA-assigned NFT. To the right of the figure, the MFSSIA DApp is shown, which comprises several components. On the one hand, MFSSIA utilizes blockchain decentralized knowledge graphs (DKG) to establish a unified view of a highly decentralized and distributed data landscape. Furthermore, MFSSIA comprises a challenge set library from which a context-dependent set of challenges are configured to be posed on a blockchain for the identity authentication of the minting context of an NFT. The responses of counterparty devices, systems, organizations, and individual humans are equally stored on a blockchain to ensure immutable traceability. To the right of Fig. 2 are the blockchain oracles depicted that deliver off-chain data for response evaluations to the challenges issued. Provided that the response evaluations match the challenges, the NFT minting is authentic.

Fig. 2 The running case of luxury item tracking with an MFSSIA secured NFT assignment [48]



Blockchain Concepts

This section provides a literature definition of the main concepts in blockchain applications commonly encountered to execute IoC. These concepts include decentralized storage, consensus mechanisms, public-key cryptography (PKC), digital signatures, smart contracts, and blockchain oracles.

There are two main types of blockchain networks: public networks and permissioned variants. Public networks allow unrestricted access to the network, while permissioned variants are private networks that restrict access to specific known parties [45]. Previous research suggests that public networks are preferred for experimenting with blockchain use cases [46]. Nevertheless, when it comes to prototyped applications that facilitate IoC, a network with permissions is the more desirable option. This preference is confirmed by the growing number of HLF network applications for DApps utilized in business collaboration scenarios [19]. HLF is a permissioned blockchain network that provides architecture and blockchain infrastructure to execute and validate the business logic contained in the chain codes in private peer-to-peer networks called channels [5]. The limited availability of blockchain applications enabling the implementation of IoC on a public blockchain can be attributed to the widespread privacy and user management challenges encountered in applications developed on public networks. Private blockchains address these concerns by implementing private data collections, which ensure the confidentiality of shared information among participants. In addition, organizations utilize certificate authorities to manage users' access to specific features by issuing and revoking certificates [25].

Decentralized storage: Data stored on blockchain networks are replicated across the nodes of parties (and organizations) that participate in collaborative processes executed on a blockchain. This ensures that there is no single point of failure, no single authority that is relied on to access data associated with a collaborative process, thus providing interoperability of data between participating organizations [46]. Due to limitations in blockchain storage spaces and associated costs, distributed data storage systems such as interplanetary file systems (IPFS) are also integrated into blockchains to address storage limitations [9]. Data stored in blockchains are organized in blocks and cryptographically linked to ensure immutability [45]. To address the challenges of scalability and efficient RWA tokenization, a new blockchain platform has been proposed, which focuses on ultra-scalable design and universal RWA tokenization [13].

Consensus mechanisms: This is an agreement mechanism for verifying, validating and adding new information represented as transactions to a blockchain network. Consensus mechanisms are largely divided into proof-based and voting-based consensus systems. In the former, a participant must obtain proof, allowing this party to aggregate and add a new

block of transactions to the network. For example, in a proof of work, the selected participant is required to have solved a difficult mathematical task. In the latter, a new block of transactions added to the network is determined based on a majority vote. Most voting-based consensus systems are adapted from Byzantine fault-tolerant systems [45].

Public key cryptography (PKC) and digital signatures: Participants in collaborative processes executed on a blockchain are identified by their key pairs; public and private keys [46]. The public key is derived from the private key and can be used to identify users. Private keys allow users to cryptographically sign transactions they publish to a blockchain network, generating a unique digital signature for each published transaction. In private blockchains such as HLF, the keys are issued by certificate authorities while they are automatically and individually generated in public networks [5, 45].

Oracles are an important bridge between a smart contract blockchain and off-chain data. Thus, with oracles, an exchange of sensitive data between smart contracts and the external environment occurs reliably and securely. The ongoing interest in nascent research in oracles [4] focuses on trust models that ensure data manipulation is prevented and data remains authentic. Notable is the "oracle problem" that addresses the challenge of ensuring that the data delivered from off-chain data sources to smart contracts are accurate and trustworthy while adhering to the decentralized and trustless characteristics of blockchains. Thus, the oracle problem is concerned with the validation and securing of smart-contract external data aligned with the blockchain security mechanisms. As a diverse spectrum of oracles evolves, the security challenges of their respective mechanisms and integration into decentralized applications must be explored to achieve greater adoption in many sectors. Interested readers are referred to [14] for more information on oracles as part of the blockchain technology stack.

Smart contracts: They are software code in the protocol layer of a blockchain network that comprises business logics. Consequently, it is possible to automatically execute specific business conditions for a collaboration context without the participation of the centralizing organizational entity. Since blockchains are closed systems, smart contracts can consequently access only data that are stored on-chain. As a remedy, the so-called oracles, being decentralized blockchain agents, are capable of transporting data between traditional off-chain databases and smart contracts [3, 46].

Tokens guide the incentive mechanisms related to a DApp by rewarding users for conducive behavior. The tokens also represent a utility value for performing the functions of the DApps [21]. Such so-called utility tokens allow for the exchange of values between IoC parties by token transfers. NFTs provide essential functions for DApps in that they are uniquely associated with RWA and digital assets

for blockchain representation [21]. Other types of tokens include identity tokens such as soul-bound tokens that are non-transferable, providing a unique representation for users in blockchain applications [50]. Soul-bound tokens (SBT) are non-transferable and a unique blockchain-user representation that can be employed for user tracking the involvement in DApp related activities. Without claiming completeness, governance tokens are yet another specific type that is usually applied in voting procedures [21].

DKG is a decentralized knowledge graph as a novel combination of blockchain technology with the semantic web, providing a decentralized, verifiable, and interoperable solution for the representation of knowledge in an IoC context [49]. The semantic web [10, 26, 41] is also termed the Web 3.0 and an extension of the Internet for adopting machine-readable data. Briefly, the Web 3.0 additions versus the current Web 2.0 are self-sovereign identity authentication and the availability of crypto wallets. This way data ownership can be clearly identity authenticated and also traded for crypto tokens. DKG involves technologies such as the resource description framework (RDF) and web ontology language (OWL) to equip data with semantics. Thereby, DKG integrates the meaning of data to be used in IoC DApps of various domains yielding an Internet with data with attached semantic meaning. The utility of knowledge graphs is extended by DKG in that blockchain's qualities of transparency, security, and immutability are leveraged to ensure the provenance and reliability of sourced knowledge. Thereby, DKG contributes to the creation of trustless DApps in which data is sourced from distributed locations for the creation of a unified view without relying on a centralized third party.

IoC Context Luxury RWA Identity Authentication

In the following, we present the process of a DApp that enables the authentication of the identity of luxury RWAs with NFTs. The stakeholders in the collaboration process comprise the manufacturer brands of the luxury RWA, the approved selling retailer, and the buying customer, who also resells the RWA. Next, we qualitatively analyze the AS-IS process for detecting security issues in the NFT minting process to also improve the configurable challenge sets.

For the remainder, Sect. [NFT-Based Authentication AS-IS Process](#) introduces the IoC AS-IS process together with NFT-based identity authentication. Next, Sect. [NFT Minting Process Security Issues](#) discusses the security issues of the NFT minting process.

NFT-Based Authentication AS-IS Process

Figure 3 depicts the authentication and verification process of the luxury RWA using NFTs in the LogisticsBDT² DApp. Note that the BPMN diagram in Fig. 3 is split into 2 stacked parts to fit a horizontal presentation in the paper, which justifies the replicated lanes with the labels *LBDT Algorithms* and *Luxury brand*. Consequently, the task with the label *Transfer Luxury NFT (update metadata)* is also duplicated to denote the continuation of the process on the BPMN split swimlanes.

The depicted process comprises four (4) process groups and four (4) process lanes, in which the first subprocess is brand registration to verify that luxury brands are responsible for creating NFTs linked to an actual physical RWA the brand produces. The second subprocess addresses NFT minting. Luxury RWAs are authenticated by the verified digital signature of the brand that is associated with the NFT genesis minting transaction. The remainder of the subprocesses specifies the NFT transfers between the brand and the retailer on the one hand, and between the retailer and the customer on the other hand. Out of focus for the current paper is the additional optional process for transferring an NFT between customers for trading pre-owned luxury RWA.

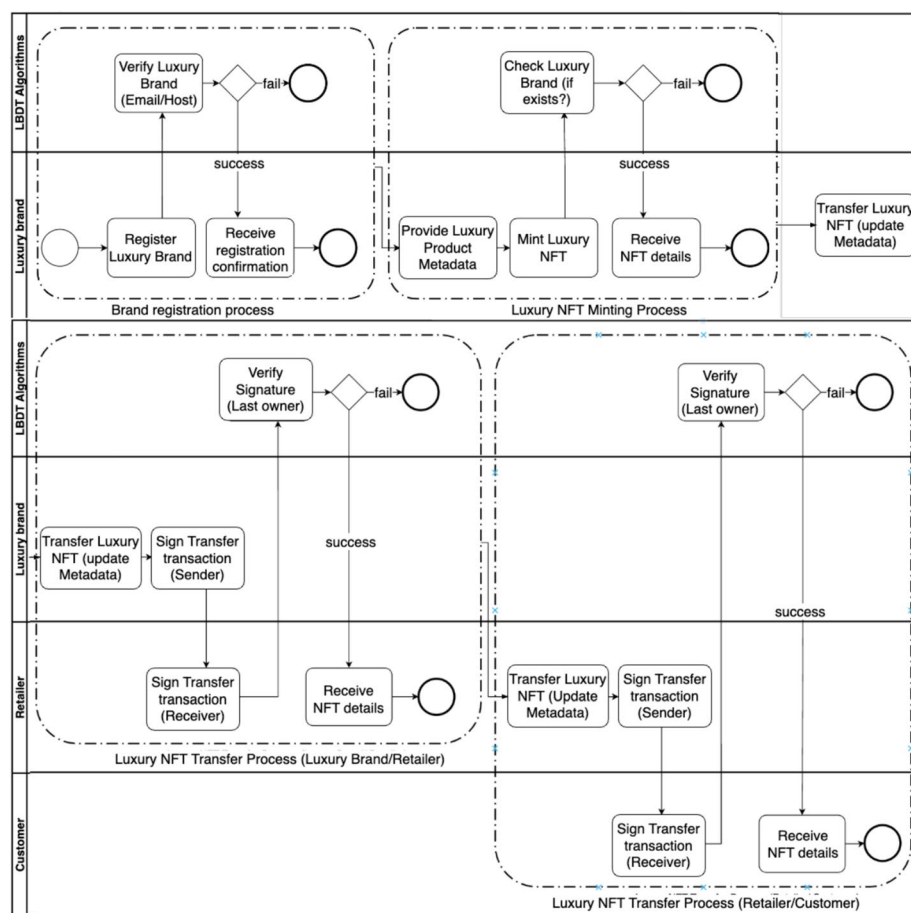
The set of conditions for the registration of a luxury brand, the minting of the corresponding NFT and the transfer of the latter, is shown in the first lane of the process in Fig. 3. The DApp algorithms contain these conditions. Luxury-specific activities are shown in the second subprocess, and comprise retailer-specific activities, while the last lane in Fig. 3 shows customer-specific activities. The DApp checks for the registration of a brand the email and host address supplied by the brand to check if it matches. In addition, a random string token is used by the DApp to assess if the authenticated identity party belongs to the specified host organization. The DApp checks for NFT minting if the minter's key is valid from a verified luxury brand. Next, the DApp checks for NFT transferral if the executor key of the function matches the last owner's key. By mapping of the distinctive ID to the NFT metadata, e.g., the serial number, product ID number, uniquely connects an NFT to a luxury RWA.

NFT Minting Process Security Issues

We identify weaknesses in the DApp's brand registration and NFT minting process with a qualitative analysis of the authentication processes. Also, other DApps that enable IoC

² LogisticsBDT luxury RWAs Authentication Platform: <https://www.logistics-bdt.com/>.

Fig. 3 AS-IS process of luxury RWA authentication with NFTs



in verifying physical/digital assets benefit from the identified issues.

- *Security weaknesses in brand verification:* Simple email and host verifications combined with a random token challenge characterize the current process. The organization can be represented on the DApp by any registered individual within that brand organization. A more robust multifactor authentication (MFA) system can be implemented to enhance brand verification security in the NFT minting process. Beyond merely considering email or host verification and a token challenge, such an MFA system requires multiple independent credentials for a trustable verification. Examples for ensuring that only authorized representatives of the brand organization can register and operate within a DApp include adding biometric verification, such as fingerprints or facial recognition. The risk of unauthorized access or representation is minimized, resulting in stronger safeguards against potential security breaches.

- *Scalability issues:* An entire organization has to rely on a single individual for digitizing an RWA with an NFT, since each organization is represented by a single key pair. This process can be completed for a smaller organization. On the other hand, multiple key pairs must be assigned to the luxury NFT minting process if larger organizations are

involved, in which specific departments and individuals focus on a single process. The introduction of a decentralized identity and access management system would address scalability issues for larger organizations in their NFT minting process. In alignment with their respective NFT minting processes' roles, such a DApp needs to allow for different key-pair distributions across different departments and individuals. In facilitating a scalable and efficient process that adapts to dynamic organizations and their requirements for digital asset management, the use of blockchain technologies ensures secure, verifiable, and revocable access rights.

- *Access control problems:* To perform specific NFT minting process tasks, user roles cannot be defined within organizations in the current brand registration process. For example, while other defined roles may perform the minting tasks, the brand registration can be completed by a role assigned to as *administrator*. Additionally, in case individuals leave a dynamic organization or change their roles, the current setup of problematic processes does not address changes that occur within an organization. A dynamic implementation of the RBAC system within the DApp could be beneficial to address access control problems. To ensure that only authorized personnel perform specific tasks, this system allows the definition of specific roles with unique permissions aligned

to the tasks in the NFT minting process. Additionally, to maintain security and operational efficiency while accommodating the dynamic organizational nature, the RBAC system could include functions for easy role reassignment and permission updates as individuals join, leave, or change roles.

We stress that the MFSSIA framework is flexible and adapts to organizational changes, e.g., for integrating new employees or the departure of existing employees. Thus, an administrator can seamlessly add a new employee joining an organization by updating the associated challenge sets. Stored and hashed on a blockchain are the generated new challenge-response pairs that reflect the novel state of the organization. On the other hand, this challenge-response pair can quickly be removed or invalidated by an administrator when an employee departs. This way, the credentials of such a former employee are no longer valid for inter-organizational process participation. The integrity and security of an organization's on-chain interactions are maintained by allowing for these updates in real-time in the MFSSIA framework. Furthermore, the MFSSIA framework integrated RBAC mechanisms are able to delineate organizational user roles for enabling designated task execution. Administrators define roles that are associated with organizational tasks. A unique set of challenge-response pairs is linked to each role. For the execution of the corresponding tasks, the challenge-response pairs must be correctly validated. Consequently, for a blockchain-enabled inter-organizational process, only authorized personnel with the required role are able to execute certain functions.

Challenge-sets for securing parties within organizations

Using organizational information, this paper proposes administered challenges to secure executed functions and parties in blockchain-based IoC. Note that information about such an organization is also dynamic, which means that only individuals have access to complete information if they are part of an organization. Thus, without hardcoding any key pair to a specific organization, functions can be executed in a collaborative business process by asking challenge questions about the organization and providing correct responses. In blockchain environments and to expand on the administrative challenges proposed, such an approach introduces a novel method to maintain security within dynamic organizational structures. By ensuring that only authorized personnel can execute specific blockchain functions, intrinsic knowledge is leveraged, being unique only to organization members. In eliminating the reliance on static key pairs, this method promotes a decentralized authentication mechanism while improving security by adapting to organizational

changes. The method allows for flexible and secure management of identities and access within blockchain systems and aligns with the principles of self-sovereign identity.

- *Organizational challenge-sets*: Fig. 4 additionally gives specific information about individuals in the organization, e.g., a passphrase. Although the responses to these questions are hashed and stored in a decentralized database, challenges are posed that represent this set of information. Any part of blockchain-based IoC is faced with challenges that must be correctly addressed. The responses are hashed and collected by an external oracle from the function-executing parties and compared to the stored on-chain hashes. To ensure that oracle operators do not receive leaked user information, oracle checks are performed confidentially. Thus, if the provided responses match the hashes stored in a decentralized database, an individual is authenticated. The number of randomly generated challenges posed depends on the level of importance of the IoC function executed. Consequently, members within an organization are integrated using this method with a sophisticated mechanism to verify and authenticate members before allowing blockchain transactions. This approach ensures that only current authorized members can initiate or approve transactions using a combination of general organizational information and individual-specific passphrases. This approach adapts to organizational changes while maintaining a secure and dynamic verification process and enhances security by leveraging unique internal knowledge of the organization. To ensure the integrity of the blockchain network, the use of an external oracle for confidential verification further protects against unauthorized access.

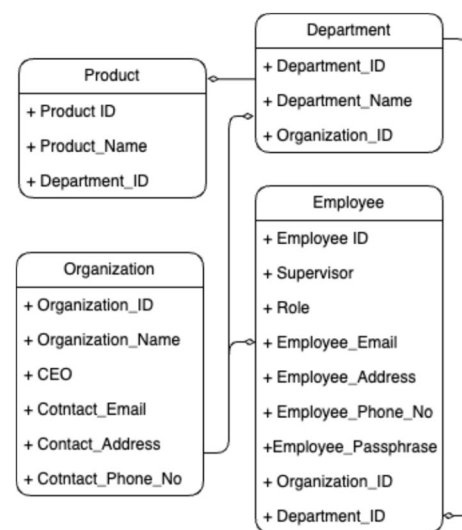


Fig. 4 Challenge-sets for verifying members of an organization

- *Administering challenges*: The administrator updates the challenges when there is a change in the state of the organization represented after their implementation by a role *administrator*. To prevent individuals who have left the organization from executing IoC functions on its behalf, the administrator also ensures regular updates to the challenge responses when there is a change in the organization's state. Furthermore, to ensure role-based access control in IoC functions, the administrator selectively deploys only hashes of challenge responses specific to certain departments or certain products. To prevent information leakage between members of organizations from different departments, passphrases are included as a mandatory part of randomly generated challenges. Each passphrase is unique for each member of an organization and is graphically linked to specific departments and roles. We ensure scalability by linking challenges to departments and products in a way that individuals working in certain departments, or products, are assigned to execute IoC functions for the organization on a blockchain. Hence, key pairs are directly linked to individuals capable of providing the correct challenges responses. To further elaborate on the administration of challenges, the role of the administrator is pivotal in maintaining the security and integrity of IoC collaborations with blockchains. The administrator plays a critical role in ensuring that only current and authorized personnel access and execute functions by dynamically updating the challenge questions and answers in response to organizational changes. Such a proactive security management approach improves the overall security posture by adapting to the evolving nature of organizational structures and roles, and prevents unauthorized access from former members.
- *Registering an admin that deploys challenges*: It is essential to present a challenge during the brand registration process that can only be successfully answered by authentic administrators of the organization given their crucial responsibility in managing and maintaining the challenges and their corresponding responses. Consequently, registered administrators deploy and update challenges for any given organization. A feasible approach is to request from an organizational administrator the provision of their host domain and email address for a random token generation, encrypted with the RSA key linked to the organizational host domain. Correctly returning the original token confirms the party as a legitimate administrator, since only the organization administrators have access to the private keys of their organization's host address. The blockchain-based system is further fortified against unauthorized access by expanding the registration process for administrators who implement challenges. This method ensures a high level of security and authentication by incorporating crypto-

graphic verification through RSA encryption of a random token linked to the organizational domain. The token can only be decrypted for completing the registration process by those with access to the private keys, e.g., legitimate administrators. For ensuring that only authorized personnel can make changes to the challenge-sets, this step adds an additional security layer and maintains the integrity and confidentiality of the organizational processes on the blockchain.

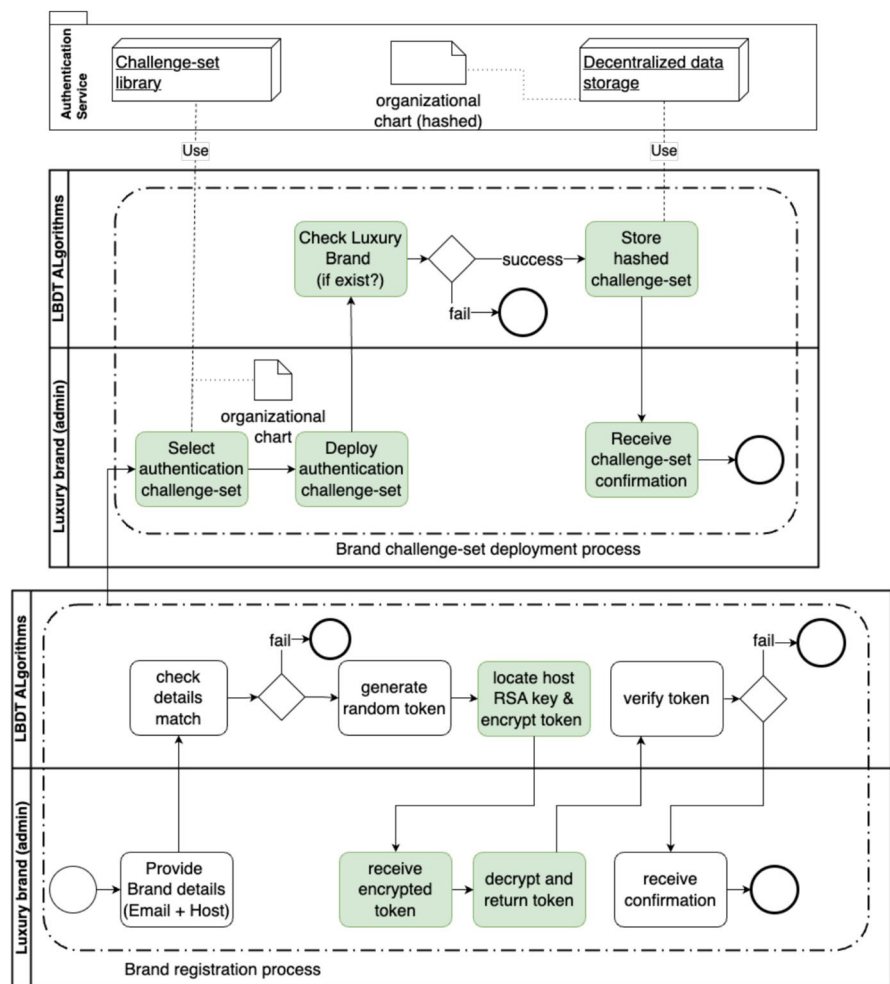
NFT-Based Luxury RWA Authentication System Challenge-Set Integration

A pivotal advance in secure digital asset management is the integration of challenge-set concepts into the NFT-based luxury RWA authentication system. This section outlines the transformative approach of incorporating dynamic, configurable challenge sets into the authentication processes for luxury RWAs, enhancing security, and ensuring only authorized transactions. By adapting the brand registration and luxury NFT minting processes, we introduce a novel method that leverages the inherent dynamism of blockchain technology, marking a significant step towards mitigating fraud and unauthorized access in the luxury goods market. Thus, we present the integration of the challenge set concepts described in the previous section into the process of authentication of luxury RWAs with NFT. In Sect. [Brand Admin Registration and Challenge-Set Deployment](#), the AS-IS process contains four subprocesses, of which only two require a challenge-set integration to improve the entire process. Thus, the TO-BE processes presented in this section contain the adapted brand registration and luxury NFT minting processes. Changes due to the integration of a configurable challenge set in the NFT authentication of the luxury RWA in Sect. [Secured Luxury NFT Minting Process](#) are discussed and marked with green process tasks.

Brand Admin Registration and Challenge-Set Deployment

Figure 5 shows the TO-BE process for brand admin registration and the deployment of configurable challenge sets for NFT minting tasks. The process started with a brand administrator providing the email and host address of the organization. The DApp algorithm checks if the host/email address matches and then proceeds to generate a random token string. The DApp algorithm also crawls the public Internet to locate the RSA public key of the host address provided. The generated random token is encrypted with the public key of the host address and sent to the email address provided by the administrator. The administrator decrypts the token to return the original token. The DApp algorithm

Fig. 5 TO-BE Admin registration and challenge-set deployment



checks if the token provided by the administrator matches the initially generated random token. The brand (admin) is registered if the token response matches the encrypted token challenge.

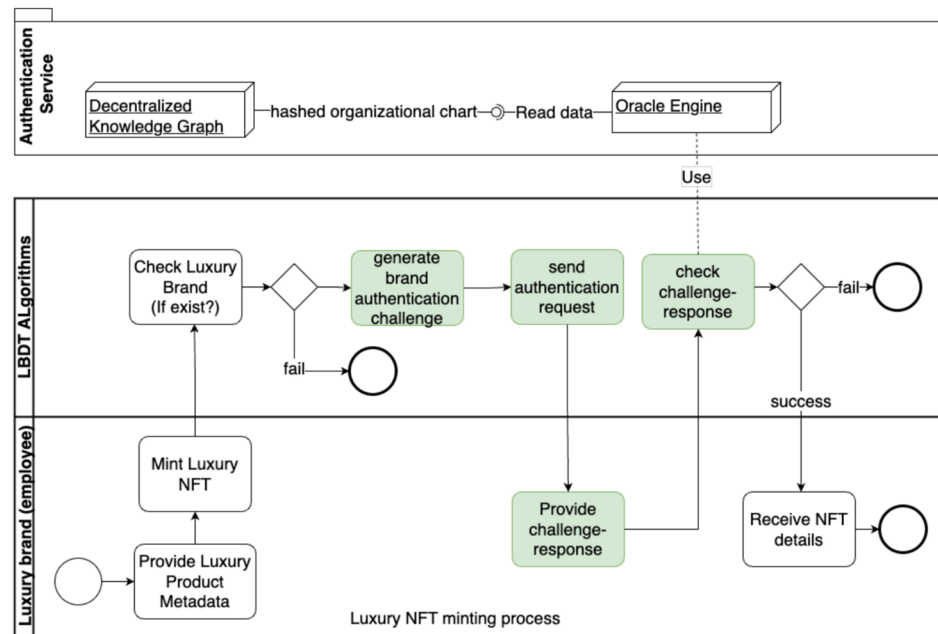
On successful registration, the admin deploys the challenge sets to authenticate the members of the registered organization before they can proceed to mint luxury NFT. To deploy the challenges, the administrator selects the relevant challenges from the challenge-set library such that the correct answers to the challenges are hashed information representing the state of the organization at the given time. The DApp algorithm checks if the key of the user deploying the challenge sets is registered as the brand administrator. If the check is positive, the challenges and the hashed responses are stored in a decentralized database that can be accessed using decentralized oracles.

Secured Luxury NFT Minting Process

Figure 6 shows the TO-BE process to produce luxury NFTs. The process starts with a representative of a luxury

brand, such as an employee, providing the details of a luxury RWA. The information is represented as NFT metadata showing the brand name, the RWA name, and other specific attributes for the RWA. A unique ID is used to digitally represent the RWA and the NFT representation is temporarily created for the RWA. The DApp algorithm checks if the brand information contained in the metadata is already a registered organization with deployed challenge sets. If the check is successful, the algorithm generates random questions based on the challenges already implemented for the employee representing the brand organization. A decentralized oracle verifies whether the response provided by the NFT minter matches the hashed data stored in the decentralized database, which represents the current state of the organization. If the response is successfully validated, the minting process for the luxury NFT is complete. Otherwise, the minting process fails.

Fig. 6 TO-BE Luxury NFT minting that includes configurable challenge sets



Evaluation and Discussion

In the evolving landscape of blockchain technology, rigorous evaluation and nuanced discussion of authentication designs are paramount to advancing secure decentralized applications. This section delves into the comparative analysis of our proposed authentication framework against existing blockchain-based technology stacks, shedding light on its innovative approach and potential implications for inter-organizational collaboration. Through a detailed examination of technology stacks and a comparative evaluation with current methodologies, we aim to underscore the significance of our contributions to the field, paving the way for enhanced security and functionality in blockchain applications. Therefore, first, Sect. [Authentication Design Technology Stacks](#) shows an assessment of the existing blockchain-based technology stacks that can be used to realize the TO-BE processes shown in Figs. 5 and 6. Then, Sect. [Authentication Design Comparative Assessment](#) provides a comparative assessment of the user authentication system presented in this work with similar works.

Authentication Design Technology Stacks

We refer to the extended MFSSIA for authenticating organizations and their parties developed in this paper as the organizational configurable challenge set system (org-ccs). The authentication design developed in this paper requires an additional technology stack, such as decentralized storage and oracles, for implementation. Some of the existing tools for the implementation of decentralized oracles include ChainLink [11] and the iExec oracle factory [23].

The Oracle iExec system is selected as the preferred choice since it provides a trusted execution environment (TEE). TEE defines an area of the processor such that tasks are executed securely without information leakage to other parts of the computing unit [44]. Therefore, tasks processed by iExec oracles are executed confidentially. Therefore, it ensures a zero-knowledge proof such that operators of decentralized oracles have no access to the authentication information provided by the users of the proposed system described in this paper.

Decentralized storage systems commonly used in blockchain applications include the Interplanetary File System (IPFS) [9] and the OriginTrail network [24]. In this paper, the OriginTrail storage system is chosen to store the challenge sets for the proposed system. This selection is based on the utilization of a decentralized knowledge graph mechanism, which facilitates the establishment of relationships between the different properties associated with the RWAs of information stored in the network. Consequently, graph-based relationships can be established for each hashed organizational chart that comprises the various subcomponents of the organizational states.

Authentication Design Comparative Assessment

In order to evaluate the authentication approach presented in this paper, we compare it to other existing solutions using a set of properties. The HLF blockchain, being a permissioned blockchain, offers a well-defined user management system that allows the implementation of authentication and access control at the organizational level [5]. Therefore, we use the HLF user management system as a basis for evaluating the

Org-CSS system developed in this paper. Furthermore, we include the SBT system as another solution for comparison, as it provides techniques to define and verify user identity properties in decentralized applications [50]. We also consider the collaborative administration of RBAC (CA-RBAC) presented in [17] as part of the related work examined. Also in [6], the authors address the issue of access control in the case of sensitive e-healthcare data. The evaluation is based on three main criteria: the presence of a certificate authority, the configurability of roles, and the flexibility of authentication approaches. The certificate authority criteria allow us to assess whether the system is capable of issuing and revoking certificates to parties or organizations involved in the decentralized application. The configurable roles criterion examines whether the system can be configured to create administrative roles and other roles within an organization participating in blockchain-based collaborative processes. Finally, the flexibility property evaluates the automation of organizations in the setup process and the ease of updating the authentication system to reflect changes within the organization.

The comparative assessment of the authentication approach designed is presented in Table 1. The HLF approach incorporates a well-defined method for issuing and revoking certificates through certificate authorities. It also offers automated onboarding of organizations using root certificates generated by member organizations. Although the authentication approach described in this paper explicitly does not involve a certificate authority, the issuance and revocation of certificates can still be achieved by an administrator who updates the hashed organizational chart. This chart contains information about specific members of the organization, ensuring that only designated members can provide accurate and valid responses to challenge questions (e.g., user passphrase) used during authentication. In the CA-RBAC system, the absence of a certificate authority is mitigated by configuring user roles in the smart contract, enabling the issuance and revocation of certificates.

In terms of keeping up with changes in organizations, the configurable challenge approach allows both the admin and the roles to be updated by providing the correct challenge responses during admin registration. However, in HLF, the

member who controls the root certificate of an organization also controls the user authentication system indefinitely. On the other hand, SBT uses a community-based approach to update and remove specific elements that define the properties of an authenticated individual. When it comes to role configurability, HLF allows administrators and other user roles to be easily specified in the policy document. In contrast, the organizational challenge-set approach only allows the administrator to be automatically configured directly, while the other roles can be indirectly specified as organizational members with their challenge-response information deployed in the authentication system. SBT offers a well-defined setup mechanism and ease of update, but currently lacks the ability to define specific roles within an organization. In CA-RBAC, the admin roles are well defined, allowing an automated voting system to administer the other roles. However, once the roles have been assigned, they cannot be updated. For initial admin roles, CA-RBAC follows a semi-automated approach.

Conclusions and Future Work

The objective of this paper is to adapt and implement a challenge-response mechanism, specifically the MFSSIA, to authenticate parties from different organizations when they engage in collaborative processes on a blockchain. The need for this study arises from the dynamic nature of organizations, which undergo frequent changes over time. Therefore, it is crucial to employ configurable and flexible authentication mechanisms to verify that parties perform IoC functions on behalf of organizations. To address this issue, we utilize MFSSIA concepts to identify appropriate authentication challenges for verifying parties and integrate a challenge-response system into an existing blockchain application that supports IoC. The paper presents the AS-IS process, which represents the blockchain-based IoC using a case study of an NFT-based luxury RWA authentication system implemented as the LogisticsBDT DApp.

The second part of the findings of this study presents a series of difficulties that arise when authenticating parties. These challenges are presented as information about the

Table 1 Assessment of authentication approaches

Authentication comparison	Property	HLF	SBT	Org-CCS	CA-RBAC
Certificates Authority	Issuing certificates	++	+	+	+
	Revoking certificates	++	+	+	+
Role Configurability	Admin roles	++	-	++	++
	Other user roles	++	-	+	+
Flexibility	Automated setup	++	++	++	+
	Ease of update	+	+	++	-

++ highly applicable, + somewhat applicable, - not applicable

present state of an organization. The correct responses are encrypted and stored on the blockchain, ensuring that only current members of the organization can provide the accurate response to each challenge. In addition, a more challenging task is used to register organizational administrators who are responsible for implementing and updating challenges to reflect any changes within the organization. The final part of the results outlines the essential steps for incorporating the developed challenge-response system into an existing DApp that is already specified in the AS-IS process.

The second half of the article focuses on evaluating and suggesting appropriate blockchain technology stacks to implement the authentication system that has been designed. The Origin Trail DKG system is proposed as a suitable option for storing challenge sets on the blockchain, while the iExec system is recommended as a reliable oracle system to securely execute the challenge-response checking mechanism. Furthermore, a comparative evaluation is conducted to compare the designed approach with other alternatives, considering factors such as the presence of a certification authority and the flexibility and configurability of the authentication systems. The authentication system that has been designed is extremely adaptable because it allows for the automatic creation of an admin and the effortless deployment of challenge sets during the onboarding process of an organization. In addition, roles in the system can be easily customized, as it includes administrators and the ability to configure challenges to accurately represent individual members or specific sections of the organization. Consequently, the framework ensures that only current and authorized employees participate in blockchain transactions by permitting easy addition or removal of employees via challenge set updates. To maintain inter-organizational process security and trustworthiness, such an adaptability is essential. This is particularly the case for dynamically changing organizational roles and personnel.

The integration of LogisticsBDT and MFSSIA DApps is not yet complete, and there is a lack of a comprehensive ontological formalization to ensure machine readability for addressing challenges and responses. To overcome these limitations, future efforts will focus on developing a proof-of-concept prototype. This prototype will enable field studies in different industry sectors, going beyond the scope of preventing counterfeit luxury item trading. With respect to the MFSSIA user-role delineation, we provide a robust mechanism for ensuring only correctly authorized employees are able to execute tasks. On the one hand, our role-based approach enhances security, and on the other hand, also ensures that organizational on-chain processes are executed efficiently and transparently on the other hand. Consequently, future work must explore additional role configuration refinements to adapt to increasingly complex organizational structures. Furthermore, investigating the

trading of audiovisual content on pay streaming platforms offers another option to conduct field research.

Author Contributions Alex Norta extended the FDSE paper to a journal paper version. Chibuzor Udokwu contributed in research design, analyses and paper writing of the original research and provided feedback on the extended journal version. Stefan Craß contributed in literature review and analyses of the original research and provided feedback on the extended journal version.

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Data Availability Statement: All data required for this research are linked to in this paper as footnotes.

Declarations

Conflict of interest The authors declare to have no conflicting interests and also no competing interests.

Research Involving Human and/or Animals This research did not involve humans and/or animals.

Informed Consent No informed consent is required as no humans are involved in this research.

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